

GEOGRAPHY

PHYSICAL GEOGRAPHY



GEOMORPHOLOGY

MODULE - 32

FLUVIAL LANDFORMS

- **EROSIONAL LANDFORMS**
- **DEPOSITIONAL LANDFORMS**

FLUVIAL LANDFORMS

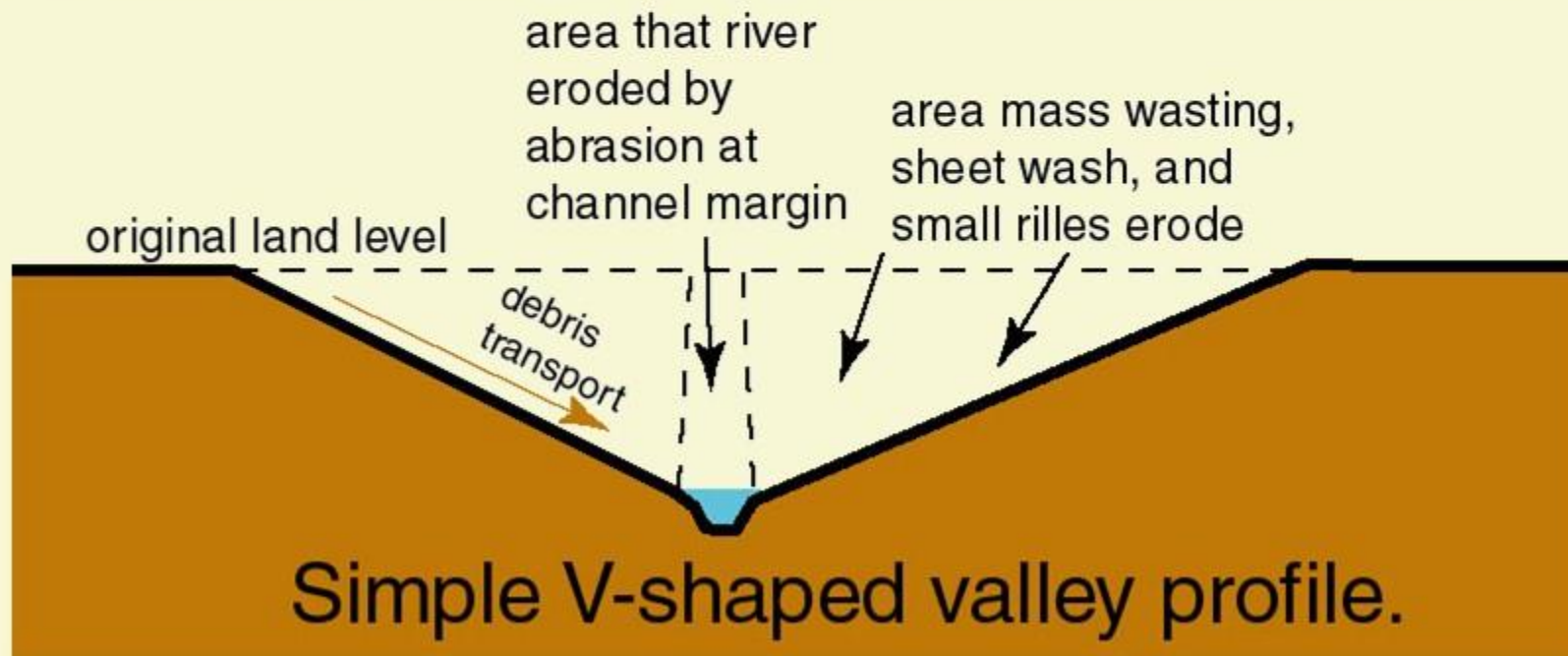
EROSIONAL LANDFORMS

1

VALLEYS

1

Valleys start as small and narrow rills, the rills will gradually develop into long and wide gullies. The Gullies will further deepen, widen and lengthen to give rise to valleys.



FLUVIAL LANDFORMS

EROSIONAL LANDFORMS

1

VALLEYS

2 Depending upon dimensions and shape, many types of valleys like V-shaped valley, gorge, canyon can be recognised.

3 A gorge is a deep valley with very steep to straight sides and a canyon is characterised by steep step-like side slopes and may be as deep as a gorge.



4 A gorge is almost equal in width at its top than at its bottom.

5 In fact, a canyon is a variant of gorge. Valley types depend upon the type and structure of rocks in which they form.

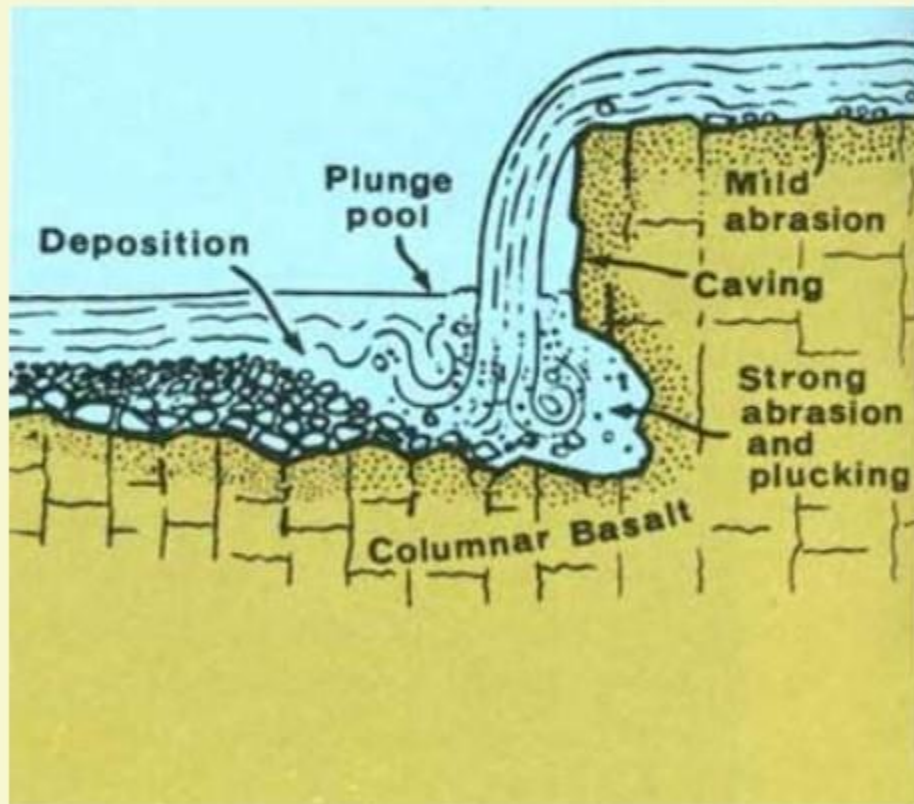


FLUVIAL LANDFORMS

EROSIONAL LANDFORMS

2

POTHOLE AND PLUNGE POOLS



1 Over the rocky beds of hill-streams **more or less circular depressions called potholes** form because of stream erosion aided by the abrasion of rock fragments.

2 Once a small and shallow depression forms, pebbles and boulders get collected in those depressions and get rotated by flowing water and consequently the depressions grow in dimensions.

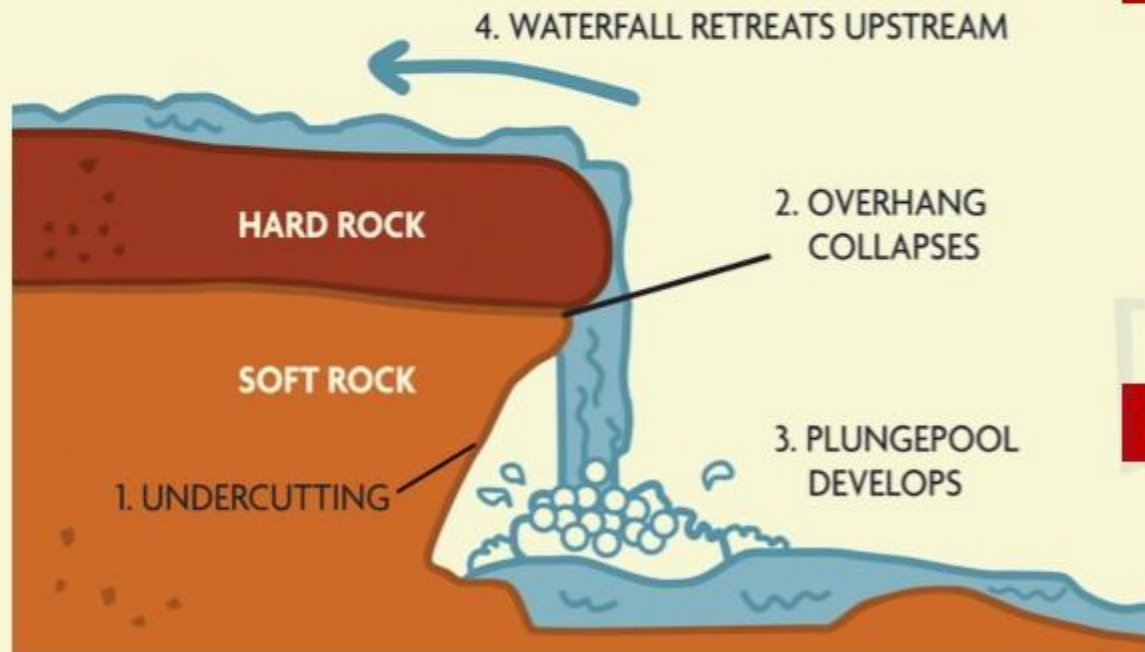
3 A series of such depressions eventually join and the stream valley gets deepened.

FLUVIAL LANDFORMS

EROSIONAL LANDFORMS

2

POTHOLE AND PLUNGE POOLS



4

At the foot of waterfalls also, large potholes, quite deep and wide, form because of the sheer impact of water and rotation of boulders. Such large and deep holes at the base of waterfalls are called **plunge pools**. These pools also help in the **deepening of valleys**.

5

Waterfalls are also transitory like any other landform and will recede gradually and bring the floor of the valley above waterfalls to the level below.

FLUVIAL LANDFORMS

EROSIONAL LANDFORMS

3 INCISED OR ENTRENCHED MEANDERS



1 In streams that flow rapidly over steep gradients, normally erosion is concentrated on the bottom of the stream channel.

2 Also, in the case of steep gradient streams, lateral erosion on the sides of the valleys is not much when compared to the streams flowing on low and gentle slopes.

3 Because of active lateral erosion, streams flowing over gentle slopes, develop sinuous or meandering courses.

FLUVIAL LANDFORMS

EROSIONAL LANDFORMS

3 INCISED OR ENTRENCHED MEANDERS

4 It is common to find meandering courses over **floodplains and delta plains** where stream gradients are very gentle.

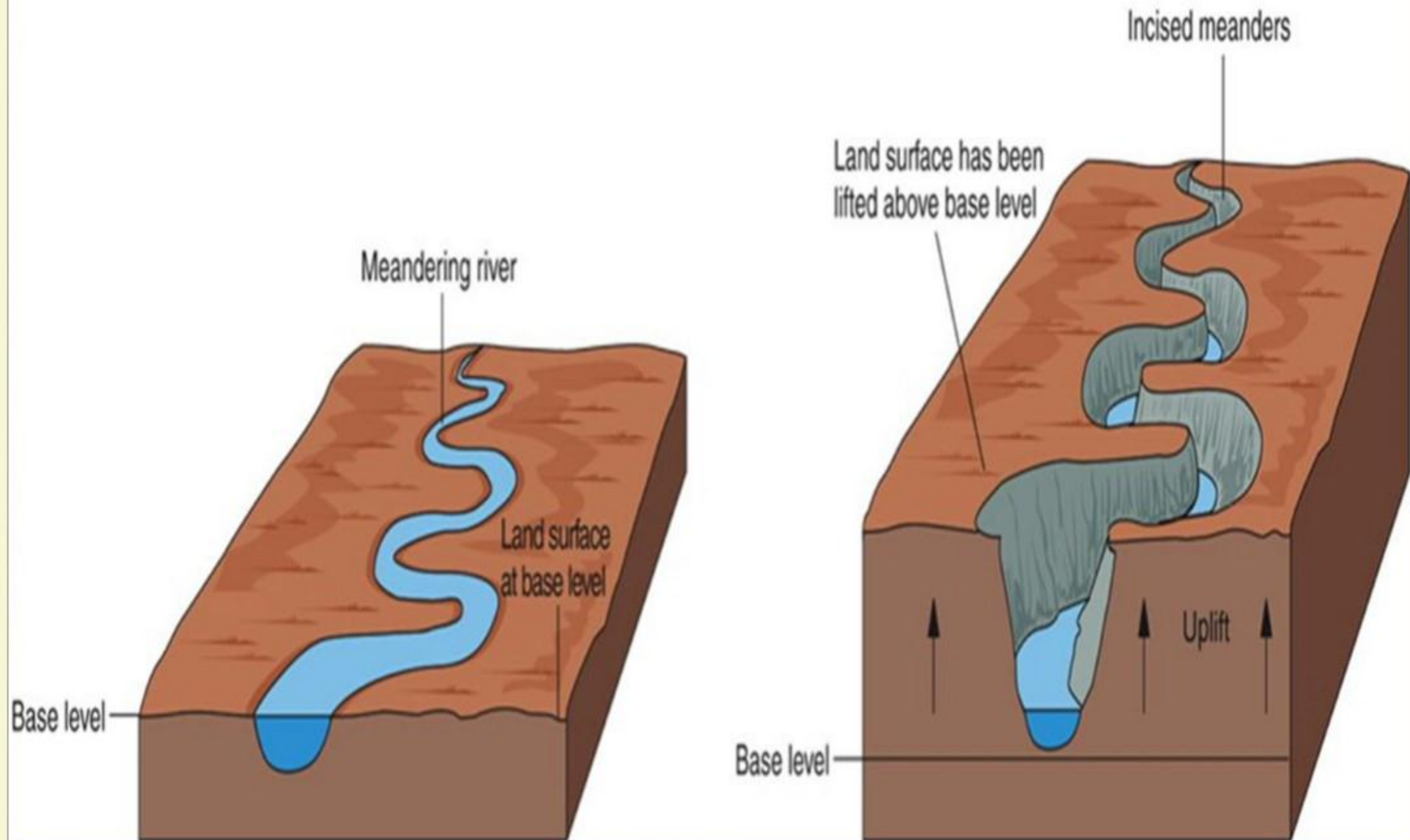
5 But very deep and wide meanders can also be found **cut in hard rocks**. Such meanders are called **Incised or Entrenched Meanders**.

6 Meander loops develop over original gentle surfaces during the development of streams.

7 The same **loops get entrenched into the rocks normally due to erosion or slow, continued uplift of the land over which they start**.

8 They widen and deepen over time and can be found as deep gorges and canyons in hard rock areas.

9 **They give an indication on the status of original land surfaces over which streams have developed.**

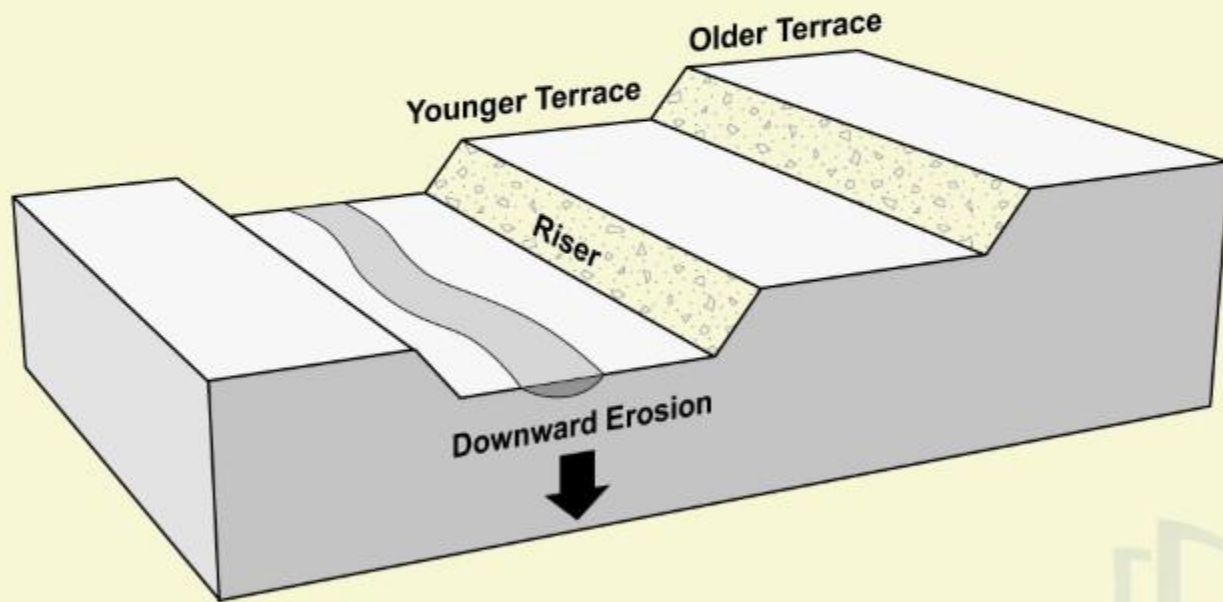


FLUVIAL LANDFORMS

EROSIONAL LANDFORMS

4

RIVER TERRACES



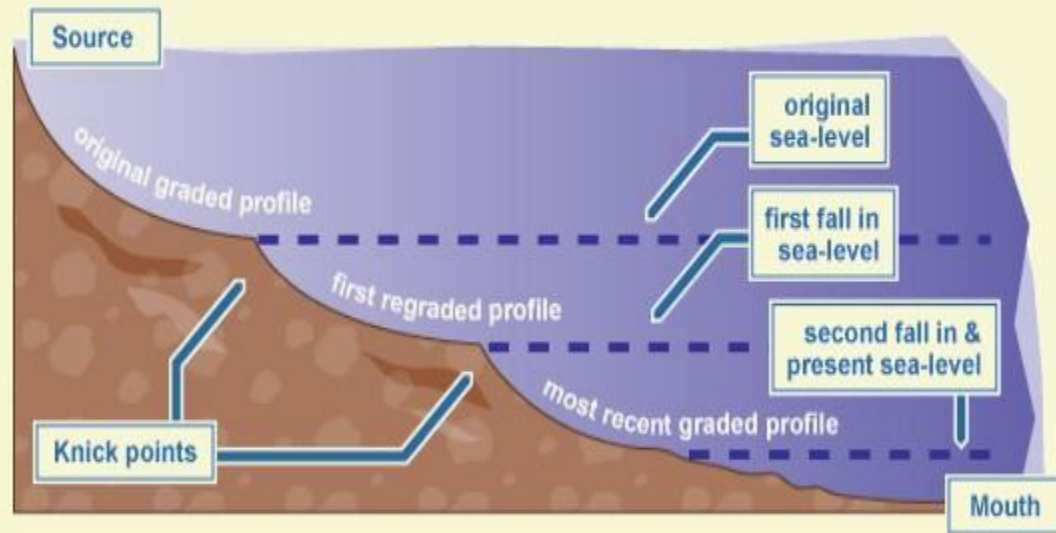
- 1 River terraces are **surfaces marking old valley floor or floodplain levels**. They may be bedrock surfaces without any alluvial cover or alluvial terraces consisting of stream deposits.
- 2 River terraces are basically products of erosion as **they result due to vertical erosion by the stream into its own depositional floodplain**.
- 3 There can be a number of such terraces at different heights indicating former river bed levels.

FLUVIAL LANDFORMS

EROSIONAL LANDFORMS

4

RIVER TERRACES



4

- The terraces may result due to
- (i) receding water after a peak flow.
 - (ii) change in hydrological regime due to climatic changes.
 - (iii) tectonic uplift of land.
 - (iv) sea level changes in case of rivers closer to sea.

FLUVIAL LANDFORMS

EROSIONAL LANDFORMS

4

RIVER TERRACES



FLUVIAL LANDFORMS

EROSIONAL LANDFORMS

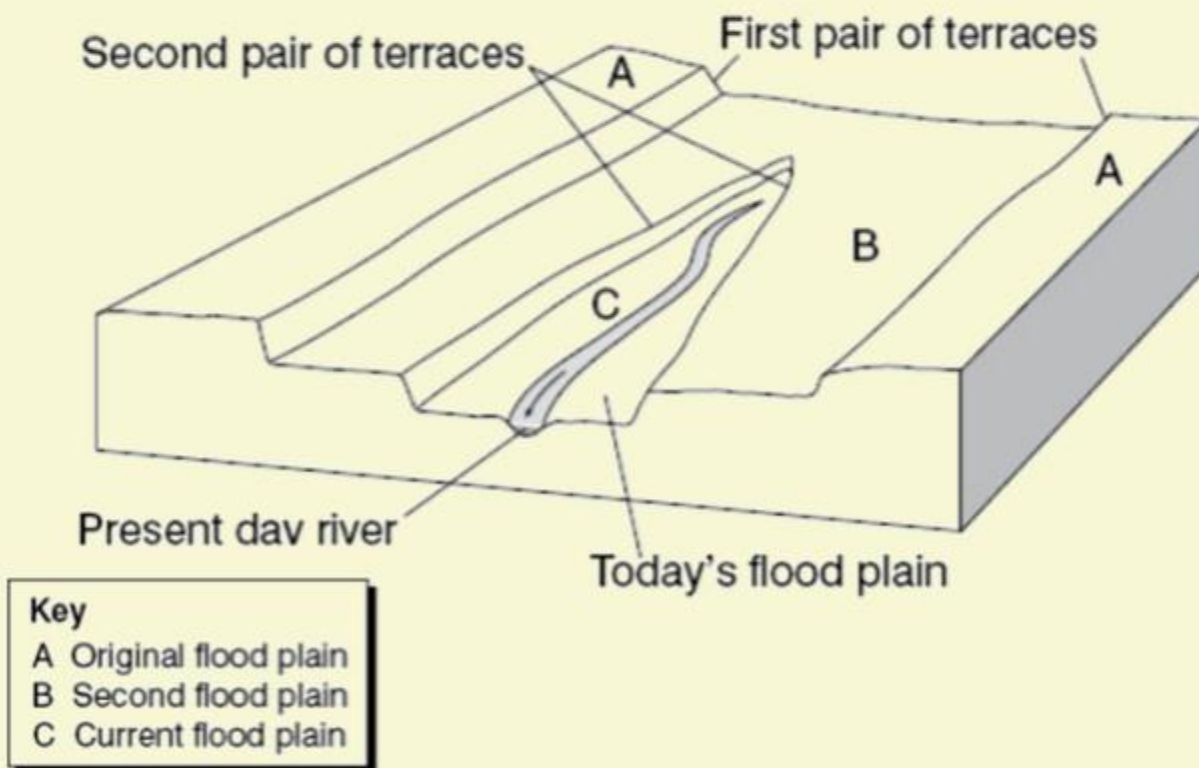
4

RIVER TERRACES

5

The river terraces may occur at the same elevation on either side of the rivers in which case they are called **paired terraces**.

Figure 3: Paired river terraces



FLUVIAL LANDFORMS

EROSIONAL LANDFORMS

4

RIVER TERRACES

Paired Terraces



Unpaired Terraces



6

When a terrace is present only on one side of the stream and with none on the other side or one at quite a different elevation on the other side, the terraces are called **unpaired terraces**.

7

Unpaired terraces are typical in **areas of slow uplift of land** or where the water column changes are not uniform along both the banks.

FLUVIAL LANDFORMS

DEPOSITIONAL LANDFORMS

1

ALLUVIAL FANS



1 Alluvial fans are formed when streams flowing from higher levels break into foot slope plains of low gradient.

2 Normally very coarse load is carried by streams flowing over mountain slopes.

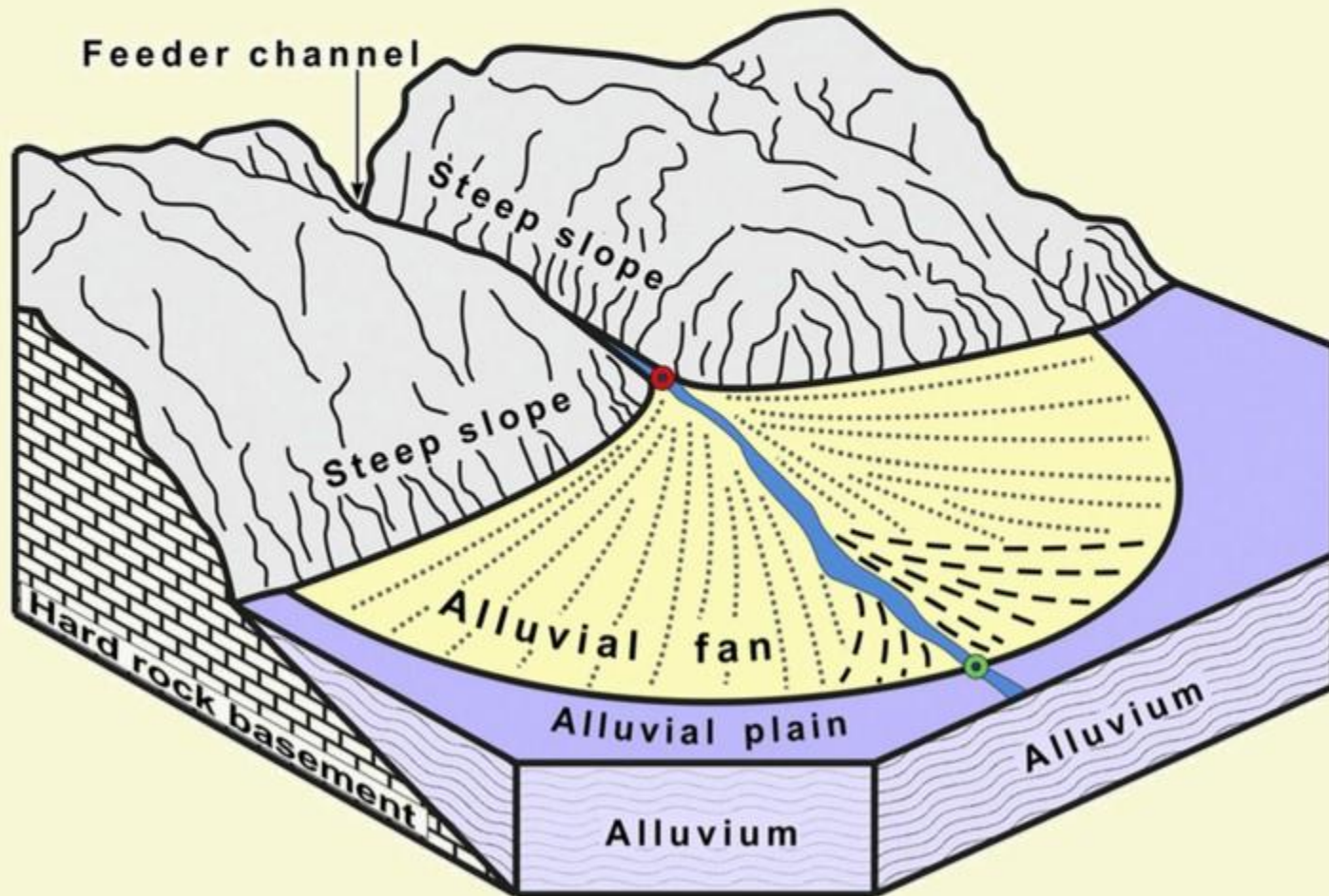
3 This **load becomes too heavy** for the streams to be carried over gentler gradients and gets dumped and spread as a broad low to high cone shaped deposit called alluvial fan.

FLUVIAL LANDFORMS

DEPOSITIONAL LANDFORMS

1

ALLUVIAL FANS



4

Usually, the streams which flow over fans are **not confined** to their original channels for long and shift their position across the fan forming many channels called distributaries.

5

Alluvial fans in **humid areas** show normally **low cones** with **gentle slope** from head to toe and they appear as **high cones** with **steep slope** in arid and semi-arid climates.

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